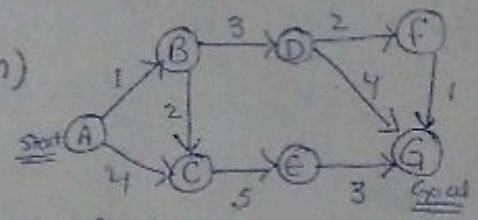
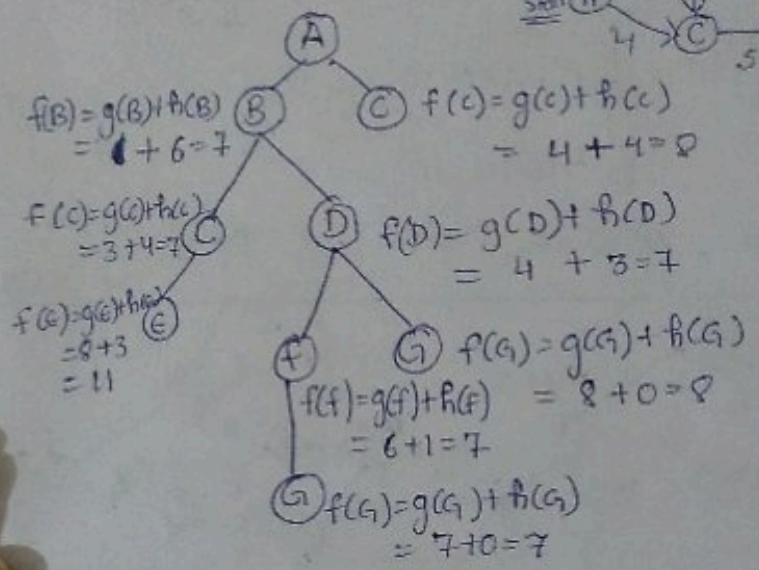
A* (A star) search

Ques 1) $f(n) = g(n) + h(n)$



A → 5
B → 6
C → 4
D → 3
E → 3
F → 1
G = 0



$f(B) = g(B) + h(B) = 1 + 6 = 7$

$f(C) = g(C) + h(C) = 4 + 4 = 8$

$f(C) = g(C) + h(C) = 3 + 4 = 7$

$f(D) = g(D) + h(D) = 4 + 3 = 7$

$f(E) = g(E) + h(E) = 8 + 3 = 11$

$f(G) = g(G) + h(G) = 8 + 0 = 8$

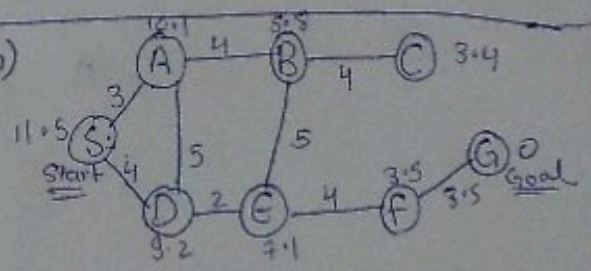
$f(F) = g(F) + h(F) = 6 + 1 = 7$

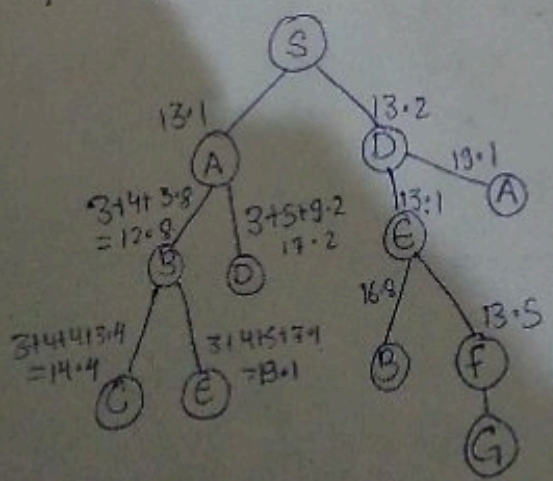
$f(G) = g(G) + h(G) = 7 + 0 = 7$

∴ path : A → B → D → F → G

path cost : 7

Ques 2) $f(n) = g(n) + h(n)$



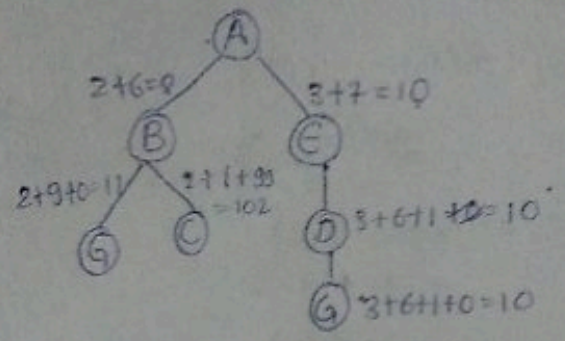


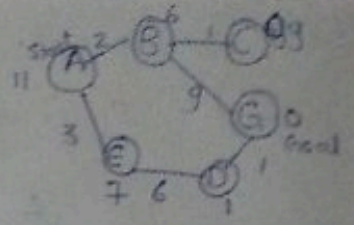
path : S → D → E → F → G

pathcost : 13.5

P(n) | debsec_gen

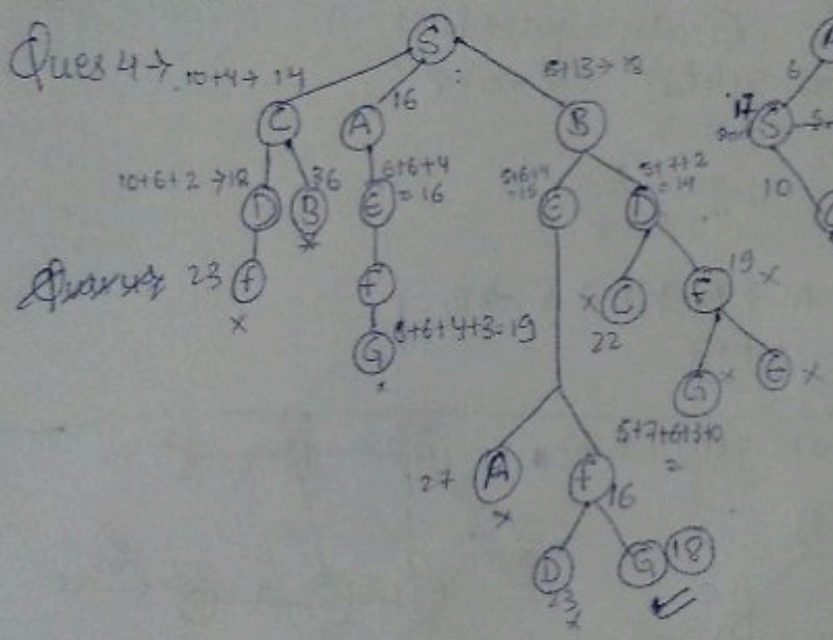
Ques 3 →

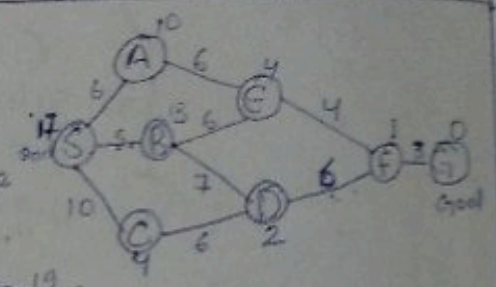




∴ path : A→E→D→G ⇒ path cost = 13

Ques 4 →

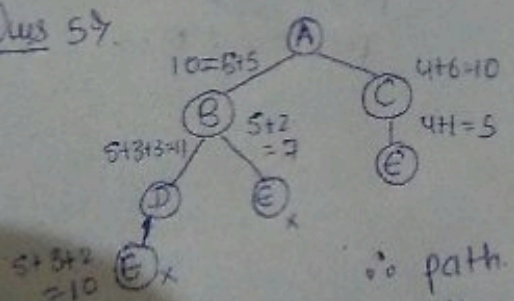


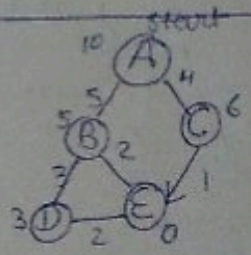


∴ path : S→B→E→F→G

path cost : 18

Ques 5 →



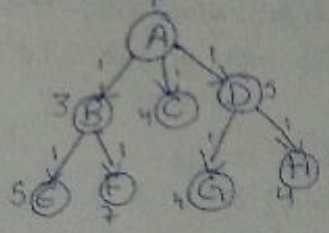


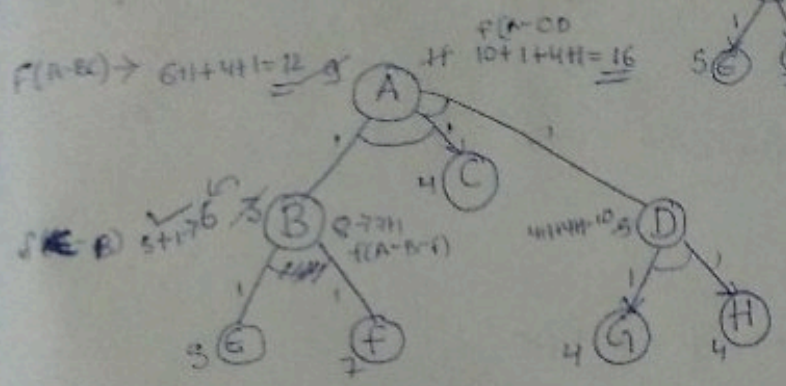
∴ path : A→C→G

path cost : 5

AO* SEARCH ALGORITHM → Solved Example

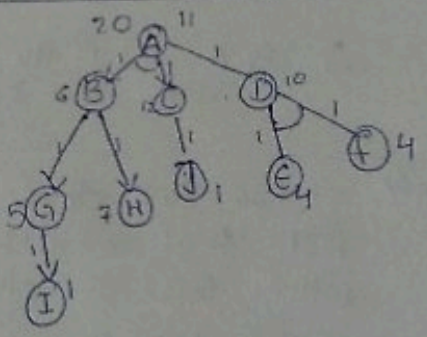
Ques ① $f(n) = g(n) + h(n)$

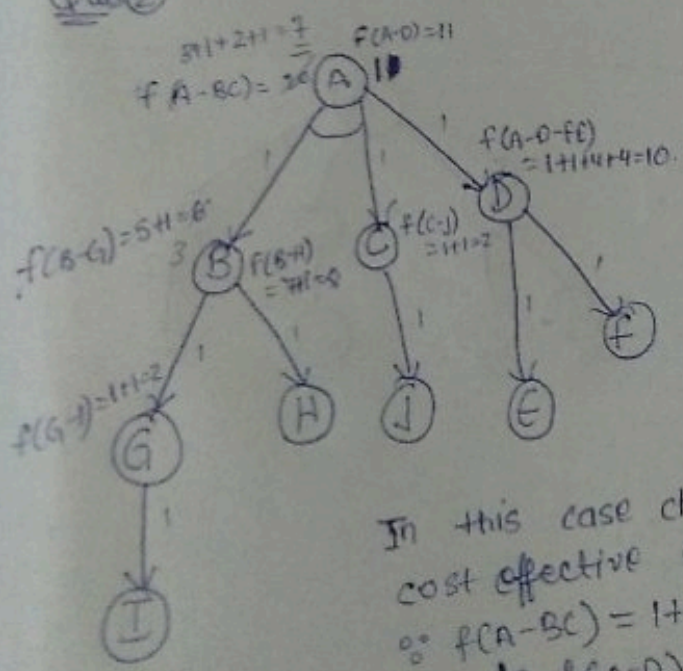




Optimal value of root not is = 12
And path considered is A-BC

Ques ②





In this case choosing path A-BC is more cost effective and good than that of A-B
∵ f(A-BC) = 1+1+2+3 = 7
while f(A-D) = 11

de bsoc_gou

f(n)

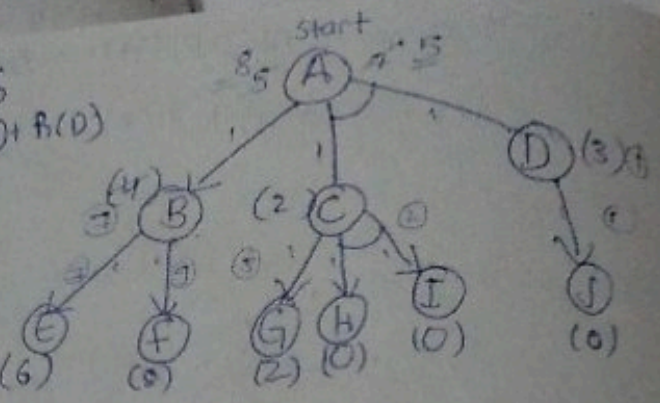
Q3 → $f(A-B) = g(B) + h(B) = 1+4+5$

$f(A-C-D) = g(C) + h(C) + g(D) + h(D)$

$= 1+2+1+3 = 7$

$f(B-E) = 1+6 = 7$

$f(B-F) = 1+8 = 9$



$f(A-B) = g(B) + \text{updated}(h(B)) = 1+7$

$= 8$

$f(C-G) = 1+2 = 3$

$f(C-H-I) = 1+0+1+0 = 2$

$f(C-H-I)$ minimum therefore cost path

$f(D-J) = 1+0 = 1$

$f(A-C-D) = g(C) + h(C) + g(D) + \text{updated}(h(D))$

$= 1+2+1+1 = 5$

∴ $f(A-CD)$ is the most cost effective path with cost → 5.

Q4 → B has two choices

path1: B-G1 cost = 0+3 = 3

path 2: B-G2 cost = 0+4 = 4

cost(B) = 1+3 = 4

[A→B = 1]

consider AND branch

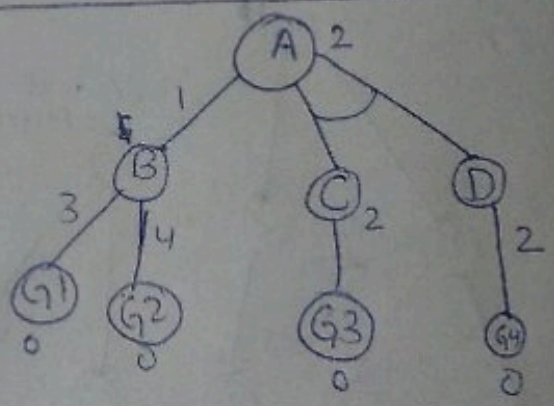
cost = 2+2+2

= 6

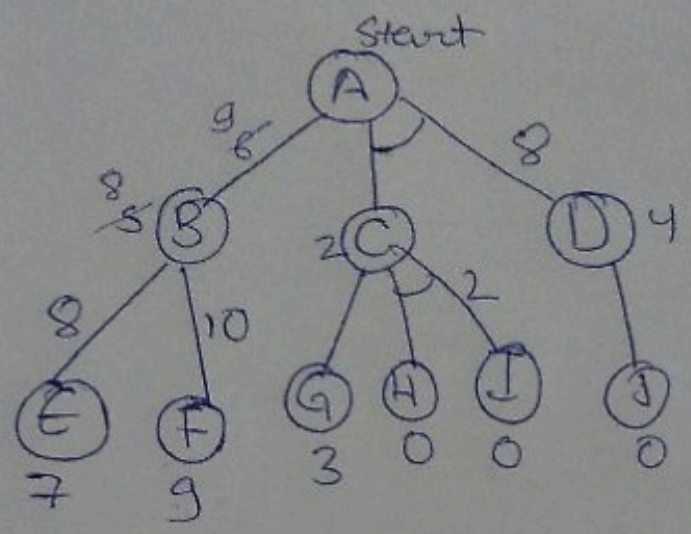
cost (AND branch) = 6

minimum is A-B-G1

$\boxed{\text{cost} = 4}$



Q5 →



∴ f(A-C-D) → is the best path with $\boxed{\text{cost} = 8}$